

## **PRACTICAL No. 4**

**AIM: To study and implement different types of SQL operators.**

**Theory:**

### **ARITHMETIC OPERATORS:**

(+): Addition - Adds values on either side of the operator.

(-): Subtraction - Subtracts right hand operand from left hand operand.

(\*): Multiplication - Multiplies values on either side of the operator.

(/): Division - Divides left hand operand by right hand operand.

(^): Power- raise to power of.

(%): Modulus - Divides left hand operand by right hand operand and returns remainder.

**Note: Write down the queries to be executed in practical slot related to arithmetic operators.**

### **LOGICAL OPERATORS:**

AND: The AND operator allows the existence of multiple conditions in an SQL statement's WHERE clause.

OR: The OR operator is used to combine multiple conditions in an SQL statement's WHERE clause.

NOT: The NOT operator reverses the meaning of the logical operator with which it is used.

Eg: NOT EXISTS, NOT BETWEEN, NOT IN, etc. This is a negate operator.

**Note: Write down the queries to be executed in practical slot related to Logical operators.**

### **COMPARISON OPERATORS:**

(=): Checks if the values of two operands are equal or not, if yes then condition becomes true.

(!=): Checks if the values of two operands are equal or not, if values are not equal then condition becomes true.

(< >): Checks if the values of two operands are equal or not, if values are not equal then condition becomes true.

(>): Checks if the value of left operand is greater than the value of right operand, if yes then condition becomes true

(<): Checks if the value of left operand is less than the value of right operand, if yes then condition becomes true.

(>=): Checks if the value of left operand is greater than or equal to the value of right operand, if yes then condition becomes true.

(<=): Checks if the value of left operand is less than or equal to the value of right operand, if yes then condition becomes true.

**Note: Write down the queries to be executed in practical slot related to Relational operators.**

### **SPECIAL OPERATOR:**

**BETWEEN:** The BETWEEN operator is used to search for values that are within a set of values, given the minimum value and the maximum value.

**IS NULL:** The NULL operator is used to compare a value with a NULL attribute value.

**ALL:** The ALL operator is used to compare a value to all values in another value set

**ANY:** The ANY operator is used to compare a value to any applicable value in the list according to the condition.

**LIKE:** The LIKE operator is used to compare a value to similar values using wildcard operators. It allows to use percent sign(%) and underscore ( \_ ) to match a given string pattern.

**IN:** The IN operator is used to compare a value to a list of literal values that have been specified.

**EXIST:** The EXISTS operator is used to search for the presence of a row in a specified table that meets certain criteria.

**Note: Write down the queries to be executed in practical slot related to Special operators.**

### **SET OPERATORS:**

The Set operator combines the result of 2 queries into a single result. The following are the operators:

- Union
- Union all
- Intersect
- Minus

**Union:** Returns all distinct rows selected by both the queries.

**Union all:** Returns all rows selected by either query including the duplicates.

**Intersect:** Returns rows selected that are common to both queries.

**Minus:** Returns all distinct rows selected by the first query and are not by the second.

**Note: Write down the queries to be executed in practical slot related to SET operators.**

### **RESULT:**

**Thus, we study and execute the different types of operators in SQL.**